



Haskell for Computational Linguists

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What are we doing today?

Introduction

Setup &
installation

Hello.hs

Phonology.hs

Syntax.hs

I will (gently) introduce you to a new paradigm, **functional programming**, and a new language, **Haskell**.

Agenda:

- 1 (10 min.) Introduction & installation
- 2 (15 min.) Recursion, thinking functionally, and Haskell syntax
- 3 (15 min.) Building a Finnish vowel harmonizer
- 4 (15 min.) Building an English → German VP translator



Who am I to tell you what to do?

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Hello y'all :) My name is Pranav. I'm a fourth-semester M.Sc. student at Universität Stuttgart, studying Computational Linguistics (specializing in Digital Phonetics).

Previously:

- **2021 - 2023:** Product Manager @ New Relic
- **2017 - 2021:** B.A. in Linguistics & Computer Science @ University of California, Los Angeles
- **1999 - 2005:** My parents' favorite child

Coding:

- **before 2017:** Python, HTML
- **since 2017:** C/C++, Java, JavaScript, Go, Haskell, OCaml, Rust



Why care about Haskell?

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- **Readable:** *Function composition* keeps code short and process-driven
- **Strong pattern-matching:** Easy to unpack and inspect custom data types
- **Elegant:** (My opinion, but) code looks clean and there's cute "syntactic sugar" everywhere

Specifically for linguists:

- **Recursion instead of iteration:** Easier to work with trees!
- **Actually used in our field:** For example, in my LING 185A: Computational Linguistics course at UCLA
 - Search "ling 185a ucla github" to find examples of previous student work, all in Haskell



What do you get out of it?

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- A new language to add to your resume!
- Finally something that isn't Python
- Friendlier build tools like `stack` and `cabal`
 - Although `uv` has made Python environment management a lot easier ;)
- Very natural syntax compared to C++, Java, and other declarative languages



What do you need today?

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- 1 Somewhere to write (VSCode, Zed...)
- 2 Somewhere to run (iTerm2, Terminal, etc...)
- 3 A free directory

If the installation does not work, you can use an online compiler, such as:

`play.haskell.org`



If you need to use an online compiler

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While I will be demonstrating code running like this...

```
→ tacos $ ghci
GHCi, version 9.4.8: https://www.haskell.org/ghc/  :? for help
ghci> :l Hello
[1 of 1] Compiling Hello                ( Hello.hs, interpreted )
Ok, one module loaded.
ghci> square 5
25
ghci> factorialList [1, 2, 3, 4, 5]
[1,2,6,24,120]
ghci> 
```



If you need to use an online compiler

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You will need to run things like this:



```
1 square :: Int -> Int
2 square x = x * x
3
4 factorial :: Int -> Int
5 factorial x
6   | x < 0    = error "(factorial) input cannot be negative"
7   | x == 0   = 1
8   | otherwise = x * (factorial (x - 1))
9
10 main :: IO ()
11 main = do
12   print (square 5)
13   print (factorial 5)
```

25
120



Installation & setup

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- 1 haskell.org/ghcup/install
- 2 “How to install” → Download GHCup



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- 1 haskell.org/ghcup/install
- 2 “How to install” → Download GHCup
- 3 Copy “...Stub.hs” files from my GitHub into your text editor
- 4 Remove the “...Stub” and edit the files directly
- 5 Test using `ghcup` in your command line



GHCi cheat sheet

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A read-evalue-print loop (REPL) for Haskell.

- 1 `$ ghci`
 - starts GHCi
- 2 `ghci> x = 5`
 - sets “x” to “5”
- 3 `ghci> x`
 - prints value of “x”
- 4 `ghci> y`
 - throws “Variable not found in scope” error
- 5 `ghci> :l Module`
 - loads file “Module.hs”
- 6 `ghci> :q`
 - quits GHCi

Let's try it!

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- 1 Create `Hello.hs`
- 2 Copy text from `HelloStub.hs` from my GitHub
- 3 Use `$ ghci` to test how you fare!

`github.com/pranavs99/tacos-workshop`



Finnish vowel harmony

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How can we model the following phenomena?

- (1) *minä asu -n tä -ssä maa -ssa*
1.SG.NOM live -1.SG PROX -INESS country -INESS

“I live this-in country-in”

“I live in this country”

- (2) *hän muista -: tuo -sta kiele -stä*
3.SG.NOM remember -3.SG DIST -ELAT language -ELAT

“they remember-s that-from language-from”

“they (sg.) remember that language”



How does Finnish vowel harmony work?

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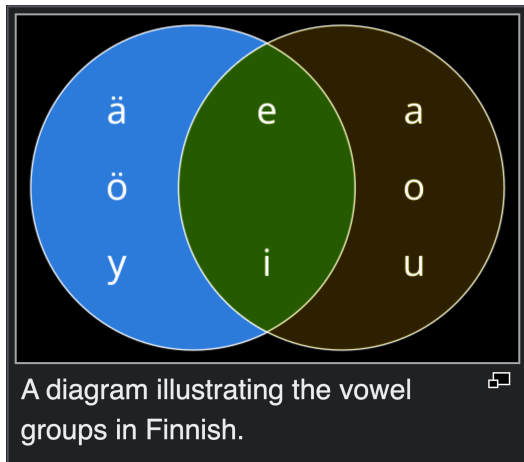
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blue = front, green = neutral, brown = back



How can we describe this as linguists?

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1 Affixes comes in two forms: **front** and **back**



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- 1 Affixes comes in two forms: **front** and **back**
- 2 Vowels in affixes **match the frontness of the stem**



How can we describe this as linguists?

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- 1 Affixes comes in two forms: **front** and **back**
- 2 Vowels in affixes **match the frontness of the stem**
- 3 When a stem only contains neutral vowels (“kiele”), the **front** affix form is used



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- 1 Affixes comes in two forms: **front** and **back**
- 2 Vowels in affixes **match the frontness of the stem**
- 3 When a stem only contains neutral vowels (“kiele”), the **front** affix form is used
 - → front form is the **default**



German V2 movement

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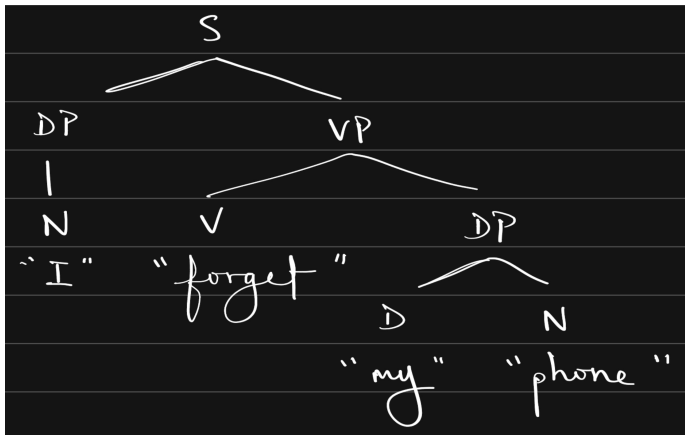
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"I forget my phone"



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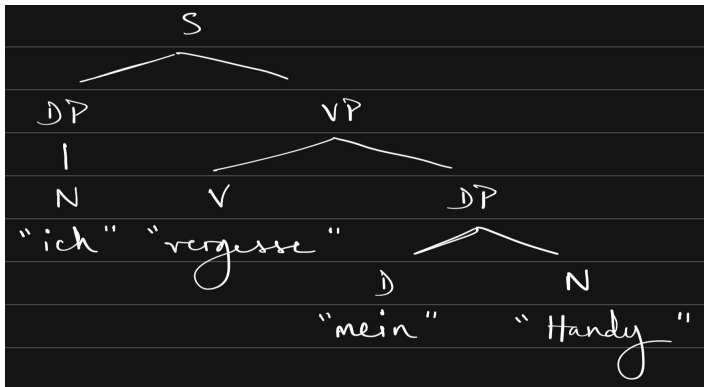
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“I forget my phone” → “ich vergesse mein Handy”



German V2 movement

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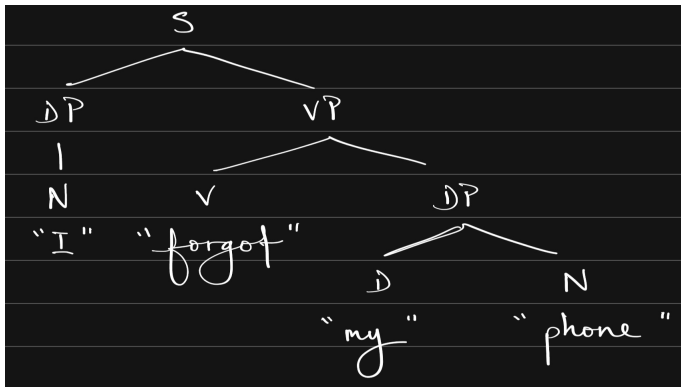
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"I forgot my phone"



German V2 movement

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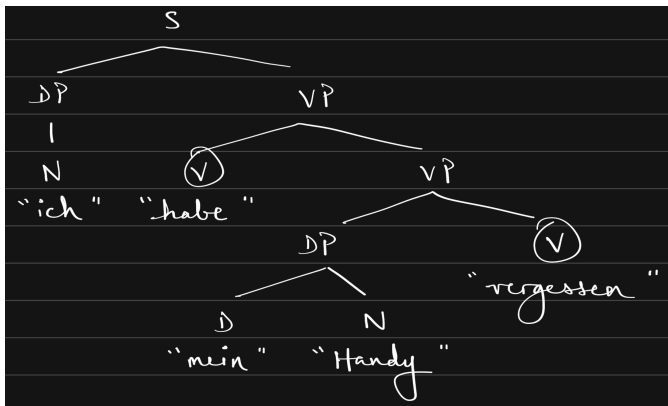
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“I forgot my phone” → “ich **habe** mein Handy **vergessen**”



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Thank you! Any questions?

